

$$\sigma : \mathcal{U} \rightarrow \mathbb{R}^3$$



$\sigma : U \rightarrow \mathbb{R}^3$ where $U \subset \mathbb{R}^2$

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$$\sigma_x$$



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 σ_x, σ_y

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Take the dot product with σ_x on both sides:

Suggestion: to find out a by taking the dot product just as we did for curves

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$$E = \sigma_x \cdot \sigma_x, F = \sigma_x \cdot \sigma_y, G = \sigma_y \cdot \sigma_y$$

This will allow us to compute for example arc lengths of curves on a surface

Consider a curve parametrized by $\theta : (a, b) \rightarrow \mathbb{R}^3$

but which lies on the surface patch

i.e. $\theta(a, b) \subset \sigma(U)$

$$\theta(t) = \sigma(x(t), y(t))$$

The arc length of a curve:

$$\int_{t_0}^{t_1} \|\theta'(t)\| dt = \int_{t_1}^{t_2} \sqrt{E(x(t), y(t))x'^2(t) + 2F(x(t), y(t))x'(t)y'(t) + G(x(t), y(t))y'^2(t)} dt$$

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$\sigma : U \rightarrow \mathbb{R}^3$ where $U \subset \mathbb{R}^2$
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$$\theta'(t) = x'(t)\sigma_x(x(t), y(t)) + y'(t)\sigma_y(x(t), y(t))$$

$$\|\theta'(t)\|^2 = \theta'(t) \cdot \theta'(t) = (x'\sigma_x + y'\sigma_y) \cdot (x'\sigma_x + y'\sigma_y)$$

$$= x'^2\sigma_x \cdot \sigma_x + 2x'y'\sigma_x \cdot \sigma_y + y'^2\sigma_y \cdot \sigma_y$$

$$= Ex'^2 + 2Fx'y' + Gy'^2$$

$E, F,$ and G together is called the *First Fundamental Form*

In general, they help us to find the dot product of tangent vectors

Supposing $\mathbf{v}$ and $\mathbf{w}$ are tangent to the surface at p

$$\mathbf{v} = a\sigma_x + b\sigma_y$$

$$\mathbf{w} = c\sigma_x + d\sigma_y$$

$$\mathbf{v} \cdot \mathbf{w} = acE + (ad + bc)F + bdG$$

Returning to:

$$\sigma_{xx} = a\sigma_x + b\sigma_y + c\mathbf{n}$$

$$\sigma_{xx} \cdot \sigma_x = a(\sigma_x \cdot \sigma_x) + b(\sigma_y \cdot \sigma_x)$$

$$\sigma_{xx} \cdot \sigma_x = aE + bF$$

Alternatively

$$\sigma_{xx} \cdot \mathbf{n} = a(\sigma_x \cdot \mathbf{n}) + b(\sigma_y \cdot \mathbf{n}) + c(\mathbf{n} \cdot \mathbf{n})$$

$$c = \sigma_{xx} \cdot \mathbf{n}$$

$$\begin{aligned}
 \theta'(t) &= x'(t)\sigma_x(x(t), y(t)) + y'(t)\sigma_y(x(t), y(t)) \\
 \|\theta'(t)\|^2 &= \theta'(t) \cdot \theta'(t) = (x'\sigma_x + y'\sigma_y) \cdot (x'\sigma_x + y'\sigma_y) \\
 &= x'^2\sigma_x \cdot \sigma_x + 2x'y'\sigma_x \cdot \sigma_y + y'^2\sigma_y \cdot \sigma_y \\
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$$\sigma_{xx} \cdot \mathbf{n} = a(\sigma_x \cdot \mathbf{n}) + b(\sigma_y \cdot \mathbf{n}) + c(\mathbf{n} \cdot \mathbf{n})$$

$$c = \sigma_{xx} \cdot \mathbf{n} =: L$$

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$$\begin{aligned}\theta'(t) &= x'(t)\sigma_x(x(t), y(t)) + y'(t)\sigma_y(x(t), y(t)) \\ \|\theta'(t)\|^2 &= \theta'(t) \cdot \theta'(t) = (x'\sigma_x + y'\sigma_y) \cdot (x'\sigma_x + y'\sigma_y) \\ &= x'^2\sigma_x \cdot \sigma_x + 2x'y'\sigma_x \cdot \sigma_y + y'^2\sigma_y \cdot \sigma_y \\ &= Ex'^2 + 2F x'y' + Gy'^2\end{aligned}$$

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$$\sigma_{xx} = a\sigma_x + b\sigma_y + c\mathbf{n}$$

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$$\sigma_{xx} \cdot \sigma_x = aE + bF$$

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Summary: Given a surface patch σ

$$\begin{aligned}\theta'(t) &= x'(t)\sigma_x(x(t), y(t)) + y'(t)\sigma_y(x(t), y(t)) \\ \|\theta'(t)\|^2 &= \theta'(t) \cdot \theta'(t) = (x'\sigma_x + y'\sigma_y) \cdot (x'\sigma_x + y'\sigma_y) \\ &= x'^2\sigma_x \cdot \sigma_x + 2x'y'\sigma_x \cdot \sigma_y + y'^2\sigma_y \cdot \sigma_y \\ &= Ex'^2 + 2Fx'y' + Gy'^2\end{aligned}$$

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$$\sigma_{xx} = a\sigma_x + b\sigma_y + c\mathbf{n}$$

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Summary: Given a surface patch σ , we compute σ_x , σ_y

$$\begin{aligned}\theta'(t) &= x'(t)\sigma_x(x(t), y(t)) + y'(t)\sigma_y(x(t), y(t)) \\ \|\theta'(t)\|^2 &= \theta'(t) \cdot \theta'(t) = (x'\sigma_x + y'\sigma_y) \cdot (x'\sigma_x + y'\sigma_y) \\ &= x'^2\sigma_x \cdot \sigma_x + 2x'y'\sigma_x \cdot \sigma_y + y'^2\sigma_y \cdot \sigma_y \\ &= Ex'^2 + 2Fx'y' + Gy'^2\end{aligned}$$

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$$\sigma_{xx} = a\sigma_x + b\sigma_y + c\mathbf{n}$$

$$\sigma_{xx} \cdot \sigma_x = a(\sigma_x \cdot \sigma_x) + b(\sigma_y \cdot \sigma_x)$$

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Summary: Given a surface patch σ , we compute σ_x , σ_y which helps us to compute $\mathbf{n}$

$$\begin{aligned}\theta'(t) &= x'(t)\sigma_x(x(t), y(t)) + y'(t)\sigma_y(x(t), y(t)) \\ \|\theta'(t)\|^2 &= \theta'(t) \cdot \theta'(t) = (x'\sigma_x + y'\sigma_y) \cdot (x'\sigma_x + y'\sigma_y) \\ &= x'^2\sigma_x \cdot \sigma_x + 2x'y'\sigma_x \cdot \sigma_y + y'^2\sigma_y \cdot \sigma_y \\ &= Ex'^2 + 2Fx'y' + Gy'^2\end{aligned}$$

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$$\sigma_{xx} \cdot \sigma_x = a(\sigma_x \cdot \sigma_x) + b(\sigma_y \cdot \sigma_x)$$

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Summary: Given a surface patch σ , we compute σ_x , σ_y which helps us to compute $\mathbf{n}$, $\sigma_x \cdot \sigma_x =: E$,

$$\begin{aligned}\theta'(t) &= x'(t)\sigma_x(x(t), y(t)) + y'(t)\sigma_y(x(t), y(t)) \\ \|\theta'(t)\|^2 &= \theta'(t) \cdot \theta'(t) = (x'\sigma_x + y'\sigma_y) \cdot (x'\sigma_x + y'\sigma_y) \\ &= x'^2\sigma_x \cdot \sigma_x + 2x'y'\sigma_x \cdot \sigma_y + y'^2\sigma_y \cdot \sigma_y \\ &= Ex'^2 + 2Fx'y' + Gy'^2\end{aligned}$$

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$$\sigma_{xx} = a\sigma_x + b\sigma_y + c\mathbf{n}$$

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$$\begin{aligned} \theta'(t) &= x'(t)\sigma_x(x(t), y(t)) + y'(t)\sigma_y(x(t), y(t)) \\ \|\theta'(t)\|^2 &= \theta'(t) \cdot \theta'(t) = (x'\sigma_x + y'\sigma_y) \cdot (x'\sigma_x + y'\sigma_y) \\ &= x'^2\sigma_x \cdot \sigma_x + 2x'y'\sigma_x \cdot \sigma_y + y'^2\sigma_y \cdot \sigma_y \\ &= Ex'^2 + 2Fx'y' + Gy'^2 \end{aligned}$$

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$$\sigma_{xx} = a\sigma_x + b\sigma_y + c\mathbf{n}$$

$$\sigma_{xx} \cdot \sigma_x = a(\sigma_x \cdot \sigma_x) + b(\sigma_y \cdot \sigma_x)$$

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Alternatively

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$$\begin{aligned} \theta'(t) &= x'(t)\sigma_x(x(t), y(t)) + y'(t)\sigma_y(x(t), y(t)) \\ \|\theta'(t)\|^2 &= \theta'(t) \cdot \theta'(t) = (x'\sigma_x + y'\sigma_y) \cdot (x'\sigma_x + y'\sigma_y) \\ &= x'^2\sigma_x \cdot \sigma_x + 2x'y'\sigma_x \cdot \sigma_y + y'^2\sigma_y \cdot \sigma_y \\ &= Ex'^2 + 2Fx'y' + Gy'^2 \end{aligned}$$

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$$\sigma_{xx} \cdot \sigma_x = a(\sigma_x \cdot \sigma_x) + b(\sigma_y \cdot \sigma_x)$$

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Alternatively

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$$\begin{aligned}\theta'(t) &= x'(t)\sigma_x(x(t), y(t)) + y'(t)\sigma_y(x(t), y(t)) \\ \|\theta'(t)\|^2 &= \theta'(t) \cdot \theta'(t) = (x'\sigma_x + y'\sigma_y) \cdot (x'\sigma_x + y'\sigma_y) \\ &= x'^2\sigma_x \cdot \sigma_x + 2x'y'\sigma_x \cdot \sigma_y + y'^2\sigma_y \cdot \sigma_y \\ &= Ex'^2 + 2Fx'y' + Gy'^2\end{aligned}$$

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$$\sigma_{xx} = a\sigma_x + b\sigma_y + L\mathbf{n}$$

$$\begin{aligned}\theta'(t) &= x'(t)\sigma_x(x(t), y(t)) + y'(t)\sigma_y(x(t), y(t)) \\ \|\theta'(t)\|^2 &= \theta'(t) \cdot \theta'(t) = (x'\sigma_x + y'\sigma_y) \cdot (x'\sigma_x + y'\sigma_y) \\ &= x'^2\sigma_x \cdot \sigma_x + 2x'y'\sigma_x \cdot \sigma_y + y'^2\sigma_y \cdot \sigma_y \\ &= Ex'^2 + 2Fx'y' + Gy'^2\end{aligned}$$

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Returning to:

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Alternatively

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Summary: Given a surface patch σ , we compute σ_x , σ_y which helps us to compute $\mathbf{n}$, $\sigma_x \cdot \sigma_x =: E$, $\sigma_x \cdot \sigma_y =: F$, $\sigma_y \cdot \sigma_y =: G$, and L , M , and N as above

$$\sigma_{xx} = a\sigma_x + b\sigma_y + L\mathbf{n}$$

$$\sigma_x \cdot \sigma_x = E$$

$$\begin{aligned}\theta'(t) &= x'(t)\sigma_x(x(t), y(t)) + y'(t)\sigma_y(x(t), y(t)) \\ \|\theta'(t)\|^2 &= \theta'(t) \cdot \theta'(t) = (x'\sigma_x + y'\sigma_y) \cdot (x'\sigma_x + y'\sigma_y) \\ &= x'^2\sigma_x \cdot \sigma_x + 2x'y'\sigma_x \cdot \sigma_y + y'^2\sigma_y \cdot \sigma_y \\ &= Ex'^2 + 2Fx'y' + Gy'^2\end{aligned}$$

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$$\mathbf{v} \cdot \mathbf{w} = acE + (ad + bc)F + bdG$$

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$$\sigma_{xx} \cdot \sigma_x = a(\sigma_x \cdot \sigma_x) + b(\sigma_y \cdot \sigma_x)$$

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Together, the form the so called *Second Fundamental Form*

Summary: Given a surface patch σ , we compute σ_x , σ_y which helps us to compute $\mathbf{n}$, $\sigma_x \cdot \sigma_x =: E$, $\sigma_x \cdot \sigma_y =: F$, $\sigma_y \cdot \sigma_y =: G$, and L , M , and N as above

$$\sigma_{xx} = a\sigma_x + b\sigma_y + L\mathbf{n}$$

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Taking partial derivatives on both sides?

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We can solve for a and b in terms of E , F , G , and their partial derivatives, and L , M , N

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$$\sigma_x \cdot \sigma_y = F$$

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$$\sigma_x \cdot \sigma_x = E$$

$$2\sigma_x \cdot \sigma_{xy} = E_y$$

$$\sigma_{xx} \cdot \sigma_y = F_x - E_y/2$$

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Taking the dot product with $\mathbf{n}$

$$L = \sigma_{xx} \cdot \mathbf{n}$$

$$M = \sigma_{xy} \cdot \mathbf{n} = \sigma_{yx} \cdot \mathbf{n}$$

$$N = \sigma_{yy} \cdot \mathbf{n}$$

Together, they form the so called *Second Fundamental Form*

Summary: Given a surface patch σ , we compute σ_x , σ_y which helps us to compute $\mathbf{n}$, $\sigma_x \cdot \sigma_x =: E$, $\sigma_x \cdot \sigma_y =: F$, $\sigma_y \cdot \sigma_y =: G$, and L , M , and N as above

$$\sigma_{xx} = a\sigma_x + b\sigma_y + L\mathbf{n}$$

$$\sigma_x \cdot \sigma_x = E$$

Taking partial derivatives on both sides?

$$2\sigma_{xx} \cdot \sigma_x = E_x$$

$$E_x/2 = aE + bF$$

$$\sigma_{xx} \cdot \sigma_y = a\sigma_x \cdot \sigma_y + b\sigma_y \cdot \sigma_y$$

$$\sigma_{xx} \cdot \sigma_y = aF + bG$$

$$\sigma_x \cdot \sigma_y = F$$

$$\sigma_{xx} \cdot \sigma_y + \sigma_x \cdot \sigma_{yx} = F_x$$

$$\sigma_x \cdot \sigma_x = E$$

$$2\sigma_x \cdot \sigma_{xy} = E_y$$

$$\sigma_{xx} \cdot \sigma_y = F_x - E_y/2$$

$$aF + bG = F_x - E_x/2$$

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We can solve for a and b in terms of E , F , G , and their partial derivatives, and L , M , N

Given any of the vectors say σ_{xx} or σ_{xy} or σ_{yy} we can express them as linear combinations of σ_x , σ_y , and $\mathbf{n}$ and their coefficients will be in terms of E , F , G , their partial derivatives, and L , M , and N

Consider the normal curvature: of a curve

$\theta : (a, b) \rightarrow \mathbb{R}^3$ on the surface

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Taking the dot product with $\mathbf{n}$

$$\theta''(t) \cdot \mathbf{n} = x'^2 L +$$

$$\sigma_{xx} \cdot \sigma_y = F_x - E_y/2$$

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Taking the dot product with $\mathbf{n}$

$$\theta''(t) \cdot \mathbf{n} = x'^2 L + 2x'y' M + y'^2 N$$

○

The main observation:

$$\sigma_{xx} \cdot \sigma_y = F_x - E_y/2$$

$$aF + bG = F_x - E_x/2$$

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$$\theta''(t) \cdot \mathbf{n} = x'^2 L + 2x'y' M + y'^2 N$$

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The normal curvature depends only on x' and y' of the curve

$$\sigma_{xx} \cdot \sigma_y = F_x - E_y/2$$

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$$\theta''(t) \cdot \mathbf{n} = x'^2 L + 2x'y' M + y'^2 N$$

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The normal curvature depends only on x' and y' of the curve. So it depends only on the velocity of the curve at that point.

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Exercise If κ_n is the normal curvature, κ_g is the geodesic curvature, and κ is the usual curvature, what is the relation between κ_n , κ_g , and κ ?

$$\sigma_{xx} \cdot \sigma_y = F_x - E_y/2$$

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